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Network Protocol Attacks Report

Introduction

Network protocols are essential for communication between systems in a network environment. However, poorly configured protocols such as FTP, Telnet, and SMB can introduce serious security vulnerabilities. These vulnerabilities may allow attackers to gain unauthorized access, intercept data, or escalate privileges.

This assessment focuses on identifying insecure network services running on the target system and demonstrating how attackers can exploit them.

Target Environment

Target Machine: Metasploitable2
Target IP Address: 192.168.206.139
Attacker Machine: Kali Linux
The testing environment consists of a vulnerable virtual machine designed for penetration testing practice and a Kali Linux machine used as the attacker system.

Network Service Discovery

The first phase of the assessment involved scanning the target system to identify open ports and running services.
Command Used:
nmap -sV 192.168.206.139
The scan revealed multiple open ports including FTP (21), Telnet (23), and SMB (445). These services are known to be vulnerable when configured improperly.
Screenshot:

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(root㉿kali)-[/home/kali]
└─# nmap -sV 192.168.206.139
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-16 13:40 -0400
Nmap scan report for 192.168.206.139
Host is up (0.0026s latency).
Not shown: 976 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rexecd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
51103/tcp open  nlockmgr    1-4 (RPC #100021)
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.25 seconds

┌──(root㉿kali)-[/home/kali]
```

FTP Vulnerability Assessment

The FTP service was tested to determine whether anonymous login was enabled. Anonymous FTP access allows attackers to connect without valid credentials.

Command Used:

ftp 192.168.206.139

The system allowed anonymous login, which indicates a misconfiguration. Attackers could potentially upload or download sensitive files through this service.

Screenshot:

```
┌──(root㉿kali)-[/home/kali]
└─# ftp 192.168.206.139
Connected to 192.168.206.139.
220 (vsFTPd 2.3.4)
Name (192.168.206.139:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Telnet Access Vulnerability

Telnet is an insecure protocol that transmits credentials in plain text. The service was tested using default credentials commonly found in vulnerable systems.

Command Used:

telnet 192.168.206.139

Credentials Used:

Username: msfadmin

Password: msfadmin

The login was successful, indicating that default credentials were still active on the system.

Screenshot:

```
┌──(root㉿kali)-[/home/kali]
└─# telnet 192.168.206.139
Trying 192.168.206.139 ...
Connected to 192.168.206.139.
Escape character is '^]'.
 _ __ ___   ___| |_ __ _ ___ _ __ | | ___ (_) |_ __ _| |__ | | ___|___ \
| '_ ` _ \ / _ \ __/ _` / __| '_ \| |/ _ \| | __/ _` | '_ \| |/ _ \ __) |
| | | | | |  __/ || (_| \__ \ |_) | | (_) | | || (_| | |_) | |  __// __/
|_| |_| |_|\___|\__\__,_|___/ .__/|_|\___/|_|\__\__,_|_.__/|_|\___|_____|
                            |_|


Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started


metasploitable login: msfadmin
Password:
Last login: Mon Mar 16 10:20:40 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
```

SMB Enumeration

The SMB service was enumerated to identify accessible network shares.
Command Used:
smbclient -L //192.168.206.139 -N
The enumeration revealed available shared directories that could potentially be accessed without authentication.
Screenshot:

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(root㉿kali)-[/home/kali]
└─# smbclient -L //192.168.206.139
Password for [WORKGROUP\root]:
Anonymous login successful

        Sharename       Type      Comment
        ---------       ----      -------
        print$          Disk      Printer Drivers
        tmp             Disk      oh noes!
        opt             Disk
        IPC$            IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
        ADMIN$          IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
Reconnecting with SMB1 for workgroup listing.
Anonymous login successful

        Server               Comment
        ---------            -------

        Workgroup            Master
        ---------            -------
        WORKGROUP            METASPLOITABLE

┌──(root㉿kali)-[/home/kali]
└─#
```

SMB Share Access

After identifying available shares, the shared directory was accessed to verify whether files could be retrieved or modified.
Command Used:
smbclient //192.168.206.139/tmp -N
Access to the shared directory was successfully established.
Screenshot:

```
┌──(root㉿kali)-[/home/kali]
└─# smbclient //192.168.206.139/tmp
Password for [WORKGROUP\root]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \>
```

Security Impact

The presence of insecure services such as FTP, Telnet, and SMB exposes the system to several security risks. Attackers can exploit these services to gain unauthorized access, steal sensitive information, or compromise the entire system.

Recommendations

1. Disable unused services such as Telnet.
2. Replace insecure protocols with secure alternatives such as SSH and SFTP.
3. Disable anonymous FTP access.
4. Implement strong authentication mechanisms.
5. Restrict SMB access through proper authentication and network segmentation.

Conclusion

This assessment demonstrates how weak network protocol configurations can lead to serious security vulnerabilities. Proper configuration, secure authentication, and regular security assessments are necessary to protect systems from unauthorized access and potential compromise.